

Chains of reasoning - Set 2 - Solutions

Specification Prf2, Prf3, Prf5.

Answer key

Question	Answer	Topic
1	E	Chains of reasoning
2	F	Chains of reasoning
3	D	Chains of reasoning
4	D	Chains of reasoning
5	H	Chains of reasoning
6	A	Chains of reasoning
7	E	Chains of reasoning
8	B	Chains of reasoning
9	E	Chains of reasoning
10	H	Chains of reasoning
11	F	Chains of reasoning

Worked solutions

Question 1 RSN-09 · Chains of reasoning · spec Prf5

A rectangle has integer side lengths and its area (in cm^2) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting $a \times b$ and $b \times a$ once)?

Answer: E

$$ab = 2a + 2b \Rightarrow (a - 2)(b - 2) = 4: (a - 2, b - 2) = (1, 4) \text{ or } (2, 2), \text{ giving } 3 \times 6 \text{ and } 4 \times 4.$$

Question 2 RSN-11 · Chains of reasoning · spec Prf5

The product of three consecutive positive integers is N , and the middle one of the three integers is m . Which of the following statements must be true?

- I N is divisible by 6.
- II N is divisible by 4.
- III $N + m$ is a perfect cube.

Answer: F

Among three consecutive integers there is a multiple of 2 and a multiple of 3 (I). II fails for $1 \cdot 2 \cdot 3 = 6$. III: $N + m = (m - 1)m(m + 1) + m = m^3 - m + m = m^3$.

Question 3 RSN-12 · Chains of reasoning · spec Prf5

How many positive integers less than 1000 are divisible by neither 2 nor 5?

Answer: D

Of 1 to 999: 499 multiples of 2, 199 of 5, 99 of 10. Divisible by 2 or 5: $499 + 199 - 99 = 599$. Neither: $999 - 599 = 400$.

Question 4 RSN-13 · Chains of reasoning · spec Prf5

p and q are prime numbers with $p + q = 45$. Find pq .

Answer: D

An odd sum of two primes needs one of them to be 2 (the only even prime): $q = 43$, which is prime. $pq = 86$.

Question 5 RSN-15 · Chains of reasoning · spec Prf5

A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?

- I The sum of all 10 terms is negative.
- II The sequence contains at least one negative term.
- III The sequence contains at least one positive term.

Answer: H

Terms 1-5 and 6-10 are two blocks of five, each with negative sum (I). A five-block has negative sum, so some term is negative (II). A three-block has positive sum, so some term is positive (III).

Question 6 RSN-17 · Chains of reasoning · spec Prf5

Alice, Ben and Cara each make one statement. Alice: 'Ben is lying.' Ben: 'Cara is lying.' Cara: 'Alice and Ben are both lying.' Exactly one of them is telling the truth. Who is it?

Answer: A

If Alice is truthful then Ben lies, so Cara is truthful: two truth-tellers, impossible. If Cara is truthful then Alice and Ben both lie; but Alice lying means Ben is truthful: contradiction. So Ben is the truth-teller: Cara lies (indeed Ben is not lying) and Alice lies (Ben is not lying), which is consistent.

Question 7 RSN-19 · Chains of reasoning · spec Prf5

A bag contains red and blue balls only. If one red ball is removed, one quarter of the remaining balls are red. If instead two blue balls are removed, one third of the remaining balls are red. How many balls are in the bag?

Answer: E

With r red and b blue: $r - 1 = \frac{1}{4}(r + b - 1)$ and $r = \frac{1}{3}(r + b - 2)$. The second gives $b = 2r + 2$; substituting into the first, $4r - 4 = 3r + 1$, so $r = 5$, $b = 12$: 17 balls.

Question 8 RSN-20 · Chains of reasoning · spec Prf5

n is a positive integer such that n^2 has exactly 3 positive divisors. Which of the following must be true?

Answer: B

A number with exactly 3 divisors is the square of a prime; so $n^2 = p^2$ and $n = p$ is prime.

Question 9 X2-26 · Chains of reasoning · spec Prf5

The positive integers a , b and c satisfy $a < b < c$ and $a + b + c = 12$. Which of the following statements must be true?

I $a \leq 3$

II $c \geq 5$

III $b = 4$

Answer: E

Since $a < b < c$, $3a < 12$ so $a \leq 3$ (I); $3c > 12$ so $c \geq 5$ (II). III fails: $1 + 2 + 9 = 12$ has $b = 2$.

Question 10 RSN-02 · Chains of reasoning · spec Prf5

The integers a , b , c satisfy $a + b + c = 0$ and $abc = 12$. Which of the following statements must be true?

I Exactly one of a , b , c is positive.

II $a^2 + b^2 + c^2$ is even.

III One of a , b , c is divisible by 3.

Answer: H

$abc > 0$ means either none or exactly two of them are negative; they cannot all be positive with sum 0, so exactly two are negative and one positive (I). $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = -2(ab + bc + ca)$ is even (II). 3 is prime and divides abc , so it divides one of them (III). (Example: $-1, -3, 4$.)

Question 11 RSN-18 · Chains of reasoning · spec Prf5

How many ordered pairs of positive integers (m, n) satisfy $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$?

Answer: F

$4n + 4m = mn \Rightarrow (m - 4)(n - 4) = 16$. Positive factor pairs of 16 as ordered pairs:

$(1, 16), (2, 8), (4, 4), (8, 2), (16, 1)$: five. (Negative factor pairs would make m or n non-positive.)